Research Article

JOURNAL OF CHEMOMETRICS

Received: 11 July 2011, Revised: 23 February 2012, Accepted: 29 February 2012, Published online in Wiley Online Library: 2012

(wileyonlinelibrary.com) DOI: 10.1002/cem.2443

Active learning for spectroscopic data regression

Fouzi Douak[a,b], Farid Melgani[a]*, Naif Alajlan[c], Edoardo Pasolli[a], Yakoub Bazi[c] and Nabil Benoudjit[b]

In this work, we introduce an active learning approach for the estimation of chemical concentrations from spectroscopic data. Its main objective is to opportunely collect training samples in such a way as to minimize the error of the regression process while minimizing the number of training samples used, and thus to reduce the costs related to training sample collection. In particular, we propose two different active learning strategies developed for regression approaches based on partial least squares regression, ridge regression, kernel ridge regression, and support vector regression. The first strategy uses a pool of regressors in order to select the samples with the greatest disagreements among the different regressors of the pool, while the second one is based on adding samples that are distant from the current training samples in the feature space. For support vector regression, a specific strategy based on the selection of the samples distant from the support vectors is proposed. Experimental results on three different real data sets are reported and discussed. Copyright © 2012 John Wiley & Sons, Ltd.

Keywords: active learning; chemical component concentration estimation; partial least squares regression (PLSR); ridge regression (RR); kernel ridge regression (KRR); support vector regression (SVR); spectroscopy

1. INTRODUCTION

Spectroscopy is an important technology for product analysis and quality control in different chemical fields. For example, it has been applied successfully in the pharmaceutical [1], [2], food [3], and textile industries [4]. Chemical analysis by spectroscopy relies on the fast acquisition of a large number of spectral data, which can be analyzed in order to yield accurate estimations of the concentration of the chemical component of interest in a given product.

From a methodological point of view, the problem of concentration estimation can be viewed as an inverse modeling issue in which it is necessary to define a model that relates the acquired observations to the concentration of interest. Typically, the model is estimated by adopting supervised regression techniques, which require the availability of a set of training samples. By training samples, we mean pairs of spectral data acquired by the spectrometer and measurements of the concentration to be estimated. In the literature, two main approaches of regression have been proposed. The first is based on linear models, appreciated for their simplicity, such as multiple linear regression, principal component regression, ridge regression (RR), and partial least squares regression (PLSR) [5]. The second approach makes use of nonlinear models. They are characterized by greater computational complexity, but they can give better performances when a strong nonlinearity between the acquired spectral data and the concentrations to be estimated is present. In this context, two state-of-the-art methods are radial basis functions neural network and support vector regression (SVR) [6], [7].

In the aforementioned works, the regression process is undertaken by assuming that the training set is composed of a sufficient number of samples in order to obtain reliable and accurate estimations. However, from a practical point of view, the process of collecting training samples is not trivial, because the concentration measurements associated with the spectral data have to be performed manually by human experts and thus are subject to errors and costs in terms of time and money. For this reason, the number of available training samples is typically limited and performances can be consequently affected owing to data scarcity. A solution to this problem is given by semi-supervised approaches [8], in which the unlabeled samples are exploited during the design of the regression model in order to compensate for the deficit in labeled samples. By unlabeled samples, we mean samples whose spectral values are known, but for which the corresponding concentration values are unknown. Such samples exhibit the advantage that they are available at zero cost from the data under analysis. In the chemometrics literature, a few regression works have been proposed in this context [9], [10].

In the data classification context, another solution to the problem of training sample collection is given by the active learning approach [11]. Starting from a small training set, additional samples are selected from a large amount of unlabeled data. These samples are labeled by the expert and added to the training set. The process is iterated until a stop criterion is reached. Active learning strategies have been applied successfully in different fields in classification [11–15].

* *Correspondence to: Farid Melgani, Department of Information Engineering and Computer Science, University of Trento, Via Sommarive, 14, I-38123, Trento, Italy.*
E-mail: melgani@disi.unitn.it

a *F. Douak, F. Melgani, E. Pasolli*
Department of Information Engineering and Computer Science, University of Trento, Via Sommarive, 14, I-38123, Trento, Italy

b *F. Douak, N. Benoudjit*
Laboratoire d'Electronique Avancée, Université de Batna, Avenue Boukhlouf Med El Hadi, 05000 Batna, Algeria

c *N. Alajlan, Y. Bazi*
ALISR Laboratory, College of Computer and Information Sciences, King Saud University, Riyadh 11543, Saudi Arabia

Copyright © 2012 John Wiley & Sons, Ltd.

Similarly, the active learning approach has been studied for regression problems by the machine learning and statistics communities, in which it is also known as *optimal experimental design*. After the seminal paper by Cohn *et al*. [16], in which active learning has been applied to two statistically based learning architectures, such as mixtures of Gaussian and locally weighted regression, several works have appeared in the last few years. For instance, in [17], the authors focus on the problem of local minima in active learning for neural networks, and two probabilistic solutions are proposed. In [18], after introducing the fundamental limits in a minimax sense of active and passive learning for various function classes, some strategies based on a tree-structured partition of the data are presented. In [19], considering linear regression scenarios, a method using the weighted least squares learning based on the conditional expectation of the generalization error is proposed. In [20], the authors apply the query by committee approach in the regression context. The main idea is to train a committee of learners and query the labels of the samples where the committee's predictions differ, thus minimizing the variance of the learner by training on samples where the variance is largest. In [21], solving the problems of active learning and model selection at the same time is suggested in order to improve further the generalization performance. In [22], a solution to the problem of pool-based active learning in linear regression is proposed. In [23], the authors develop a strategy for kernel-based linear regression, in which the proposed greedy algorithm employs a minimum-entropy criterion, derived by using a Bayesian interpretation of RR.

Despite the promising performance achieved by the active learning approach in various regression fields, nothing similar has been proposed in the chemometrics literature. Note that whereas semi-supervised methods integrate unlabeled samples automatically (without the intervention of human experts) in the learning process, active learning works differently. Indeed, its aim is to minimize the number of unlabeled samples to be labeled by human experts. For such a purpose, it resorts to smart strategies for selecting the most significant unlabeled samples, that is, those which, if labeled, would most improve the classification/regression model.

The objective of this paper is to introduce the active learning approach for regression problems for the estimation of concentrations from spectroscopic data. In particular, we propose two different active learning strategies. The first strategy uses a pool of regressors in order to select the samples with the greatest disagreements among the different regressors of the pool, while the second one is based on adding samples that are distant from the current training samples in the feature space. These strategies are tested on different linear and nonlinear methods, namely PLSR, RR, kernel ridge regression (KRR), and SVR. For SVR, a specific strategy based on the selection of the samples distant from the current support vectors is proposed. To illustrate the capabilities of the proposed strategies, we conducted an experimental study based on three different real data sets: (i) a diesel data set for estimating the cetane number by near-infrared spectroscopy; (ii) an orange juice data set where near-infrared reflectance spectroscopy is used to estimate the saccharose concentration; (iii) a Tecator data set for the estimation of fat content in meat by near-infrared transmittance spectroscopy. The obtained results show that interesting performances can be achieved.

The remaining part of the paper is organized as follows. In Section 2, the basic concepts of two linear and two nonlinear regression methods, being, PLSR, RR, KRR, and SVR, respectively, are recalled. In Section 3, the active learning strategies proposed for regression problems are described. Section 4 presents the data sets used in the experimental analysis and the related results. Finally, conclusions are drawn in Section 5.

2. LINEAR AND NONLINEAR REGRESSION TOOLS

2.1. Partial least squares and ridge regression

2.1.1. *Partial least squares regression*

Let us consider a set of labeled samples $\mathrm{L} = \{\mathbf{x}_l, y_l\}_{l=1}^{n}$, where $\mathbf{x}_l = [x_{l1}, \ldots, x_{ld}] \in \Re^d$ represents a vector of d spectral acquisitions and/or processed features and $y_l \in \Re$ is the associated target value, that is, the measurement of the concentration value of interest. Let us aggregate all $\mathbf{x}_l$'s ($l = 1, \ldots, n$) into a $n \times d$ feature matrix X and all y_l's ($l = 1, \ldots, n$) into a target vector $\mathbf{y}$ so that $\mathrm{L} = \{\mathrm{X}, \mathbf{y}\}$. The goal is to infer from the set of labeled samples L the function $f(\cdot)$ so that $\mathrm{y} = f(\mathbf{x})$.

Partial least squares regression aims to find a linear regression model by projecting data to a new space [24]. In particular, it tries to find the multidimensional direction in the space X that explains the maximum multidimensional variance direction in the space y. The user has to supply the number l_v of latent factors in the regression. If l_v equals the rank of the matrix X, the method yields simply the least squares regression estimates.

2.1.2. *Ridge regression*

Assuming that the linear regression model is expressed as:

$$\mathbf{y} = \mathrm{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} \tag{1}$$

where $\boldsymbol{\beta}$ is a vector of d regression coefficients and $\boldsymbol{\varepsilon}$ is a zero-mean Gaussian noise, the ordinary least squares estimate of $\boldsymbol{\beta}$ is given by:

$$\hat{\boldsymbol{\beta}} = \left(\mathrm{X}'\mathrm{X}\right)^{-1}\mathrm{X}'\mathbf{y} \tag{2}$$

Because of the high degree of collinearity as d increases, $\mathrm{X}'\mathrm{X}$ tends to be singular, and the estimates become unstable, thus leading to an inappropriate regression model. In order to solve this problem, one can resort to RR for which:

$$\hat{\boldsymbol{\beta}}_{ridge} = \left(\mathrm{X}'\mathrm{X} + \lambda\mathbf{I}\right)^{-1}\mathrm{X}'\mathbf{y} \tag{3}$$

where $\mathbf{I}$ is the identity matrix and λ is the ridge parameter ($\lambda \in \Re, \lambda \geq 0$) [25].

2.2. Kernel ridge regression and support vector regression

2.2.1. *Kernel ridge regression*

In RR, the prediction function takes the following dual representation:

$$f(\mathbf{x}) = \mathbf{z}'\left(\mathrm{X}\mathrm{X}' + \lambda\mathbf{I}\right)^{-1}\mathbf{y} \tag{4}$$

where $\mathbf{z}$ is the vector of z_i's such that $z_i = <\mathbf{x} \cdot \mathbf{x}_i>$, $i = 1, \ldots, n$.

Kernel ridge regression is a nonlinear form of RR, which exploits the so-called kernel trick, namely a nonlinear kernel function in the original space defining an inner product in a transformed space of higher dimensionality $\mathrm{K}_{ij} = \langle \boldsymbol{\varphi}(\mathbf{x}_i) \cdot \boldsymbol{\varphi}(\mathbf{x}_j) \rangle$

Copyright © 2012 John Wiley & Sons, Ltd.

($\mathbf{x}_i$ and $\mathbf{x}_j$ are two generic points of the original space and $\varphi(\mathbf{x}_i)$ and $\varphi(\mathbf{x}_j)$ their respective transforms) [26]. The KRR prediction function is given by:

$$f(\mathbf{x}) = \mathbf{k}'(\mathrm{K} + \lambda\mathbf{I})^{-1}\mathbf{y} \tag{5}$$

where K is the so-called Gram matrix with entries $K_{ij} = \langle \boldsymbol{\varphi}(\mathbf{x}_i) \cdot \boldsymbol{\varphi}(\mathbf{x}_j) \rangle$, and $\mathbf{k}$ is a vector with entries $k_i = < \varphi(\mathbf{x})\ \varphi(\mathbf{x}_i) > = k(\mathbf{x}, \mathbf{x}_i)$.

2.2.2. *Support vector regression*

The ε-insensitive SVR technique is based on the idea of finding an estimate $\hat{f}(\mathbf{x})$ of the true and unknown relationship $y = f(\mathbf{x})$ between the vector of spectral data $\mathbf{x}$ and the desired concentration value y from the given set of training samples L such that: (i) $\hat{f}(\mathbf{x})$ has, at most, ε deviation from the desired targets y_l ($l = 1, \ldots, n$), and (ii) it is as smooth as possible [26], [27]. This is usually performed by mapping the data from the original d-dimensional feature space to a higher-dimensional transformed feature space, that is, $\Phi(\mathbf{x}) \in \Re^{d'}(d' > d)$, to increase the flatness of the function and, accordingly, to approximate it in a linear way as follows:

$$\hat{f}(\mathbf{x}) = \omega^* \cdot \Phi(\mathbf{x}) + \mathrm{b}^*. \tag{6}$$

The SVR prediction function takes the form:

$$\hat{f}(\mathbf{x}) = \sum_{l \in S} (\alpha_l - \alpha_l^*)\ \mathrm{k}(\mathbf{x}_l, \mathbf{x}) + \mathrm{b}^* \tag{7}$$

where $\mathrm{k}(\cdot,\cdot)$ is a kernel function (a common example is the Gaussian function). S is the subset of indices ($l = 1, 2, \ldots, n$) corresponding to the nonzero Lagrange multipliers α_l's or α_l^*'s.

3. PROPOSED ACTIVE LEARNING METHODS

Let us consider a training set composed initially of n-labeled samples $\mathrm{L} = \{\mathbf{x}_l, y_l\}_{l=1}^{n}$ and an additional learning set composed of m-unlabeled samples $\mathrm{U} = \{\mathbf{x}_u\}_{u=n+1}^{n+m}$, with $m >> n$. In order to increase the training set L with a series of samples chosen from the learning set U and labeled by the expert, an active learning algorithm has the task of choosing them properly, so that the error of the regression process is minimized while minimizing the number of learning samples to be labeled, and thus to reduce the costs related to training sample collection.

In Figure 1, we show the generic flow chart of the active learning approach for regression problems proposed in this work. Starting from the initial and small training set L, the unlabeled samples of the learning set U are evaluated and sorted using an opportune criterion f_{AL}. In particular, we suppose for convention that the criterion f_{AL} has to be minimized. At this point, from the sorted samples U_s, the first n_s samples are selected, where n_s is the number of samples to be added in the training set L. Finally, the selected samples U'_s are labeled by the human expert and added to the training set L. The entire process is iterated until the predefined convergence condition is satisfied (e.g., the total number of samples to add to the training set is reached, or the accuracy improvement on an independent calibration/validation set over the last iterations becomes insignificant).

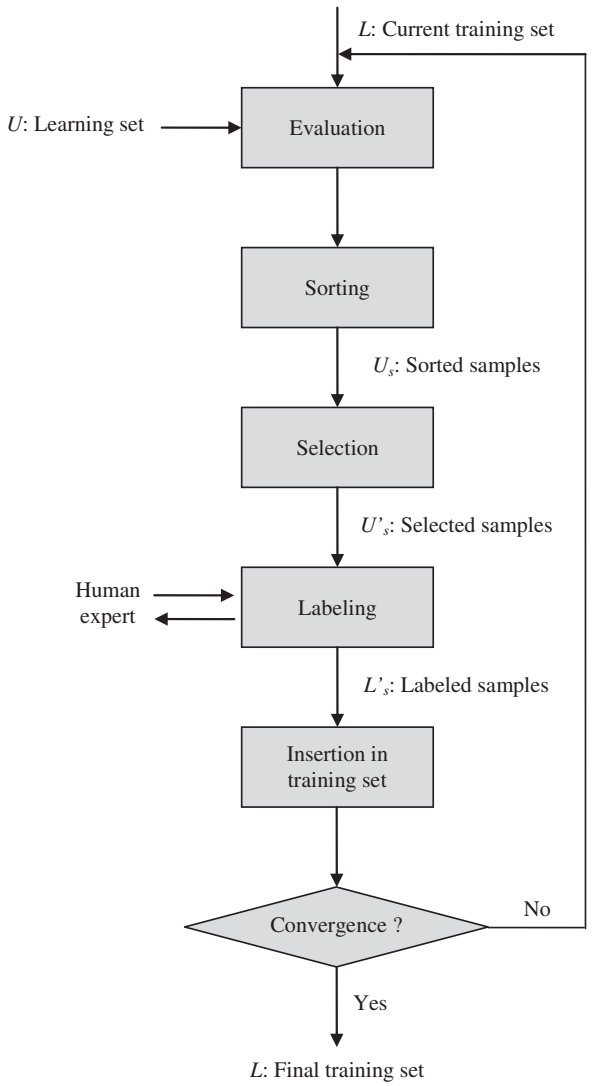


Figure 1. Flow chart of the proposed active learning approach for regression problems.

Pseudocode 1 resumes the active learning approach for regression problems.

Pseudocode 1: Active learning approach
In the following subsections, we present the different active learning strategies proposed in this work
Inputs:
L, initial training set, composed of n-labeled samples.
U, learning set, composed of m ($m >> n$) unlabeled samples.
n_s, number of samples to add at every iteration of the active learning process.
Repeat
(1) Sort the learning set U using the criterion f_{AL} in order to obtain the set U_s.
(2) Select the first n_s samples from U_s.
(3) Label the selected samples U'_s.
(4) Add the labeled samples L'_s to the training set L and remove them from U.
Until the predefined convergence condition is satisfied.
Output:
L, Final training set.

Copyright © 2012 John Wiley & Sons, Ltd.

3.1. Proposed general active learning strategies

3.1.1. Pool of regressors. The first strategy, named PAL in the rest of the paper, is based on a pool of regressors, being, PLSR, RR, or KRR depending on the adopted regression method. Considering the original set of spectral data $X = \{\mathbf{x}^1, \mathbf{x}^2, \ldots, \mathbf{x}^d\}$, q training subsets are constructed by sampling regularly the spectral domain with a step of q that each subset yielded from a different starting point of the spectrum. For instance, if $q = 2$ and assuming that d is even, the first spectral subset is $\{\mathbf{x}^1, \mathbf{x}^3, \mathbf{x}^5, \ldots, \mathbf{x}^{d-1}\}$ whereas the second is $\{\mathbf{x}^2, \mathbf{x}^4, \mathbf{x}^6, \ldots, \mathbf{x}^d\}$. Each training subset is considered independently from the others and used to train a different regressor. In this way, q parallel regressors are used. Therefore, q different estimations are obtained for each sample. Let $\mu_{u,\theta}$ be the prediction value yielded for sample $\mathbf{x}_u$ $(u = n+1, n+2, \ldots, n+m)$ by regressor r_θ $(\theta = 1, 2, \ldots, q)$. For each sample, the predictions are exploited to calculate the variance value on them:

$$f_{\mathrm{PAL},u} = \frac{1}{q}\sum_{\theta=1}^{q}\left(\mu_{u,\theta} - \bar{\mu}_u\right)^2 \tag{8}$$

where

$$\bar{\mu}_u = \frac{1}{q}\sum_{\theta=1}^{q}\mu_{u,\theta}. \tag{9}$$

The samples characterized by the greatest disagreements among the different regressors, that is, the greatest variance values, are selected. Indeed, a high disagreement means that the corresponding sample has been estimated with high uncertainty, and thus adding it to the training set could be useful in improving the regression process.

Pseudocode 2 resumes the proposed methodology based on the pool of regressors.

Pseudocode 2: Active learning based on pool of regressors
Inputs:
L, initial training set.
U, learning set.
q, number of training subsets.
n_s, number of samples to add at every iteration of the active learning process.
(1) Considering the current training set L, construct q different training subsets L_g $(g = 1, 2, \ldots, q)$ by sampling the spectral domain.
(2) Predict the target value of each sample $\mathbf{x}_u$ $(u = n+1, n+2, \ldots, n+m)$ of the learning set U for each regressor r_θ $(\theta = 1, 2, \ldots, q)$.
(3) Compute the variance on the predictions $f_{PAL,u}$ given by the different regressors using Equation (8).
(4) Set $f_{PAL}(u) = -f_{PAL,u}$
Output:
L'_s, new subset of labeled samples to add in the training set.

3.1.2. Distance from the closest training sample. The second strategy (DAL) consists of calculating for each sample $\mathbf{x}_u$ $(u = n+1, n+2, \ldots, n+m)$ the Euclidean distances $\mathbf{D}_u \in \Re^n = [D_{u,1}, D_{u,2}, \ldots, D_{u,n}]$ in the feature domain from the samples $\mathbf{x}_l$ $(l = 1, 2, \ldots, n)$ already composing the current training set:

$$D_{u,l} = \|\mathbf{x}_u - \mathbf{x}_l\|. \tag{10}$$

After that, for each sample, $\mathbf{x}_u$ $(u = n+1, n+2, \ldots, n+m)$, the closest training sample $T_{\min,u}$ is identified and the corresponding distance value $D_{\min,u}$ is considered to define the selection criterion:

$$f_{\mathrm{DAL}}(u) = -D_{\min,u} = -\min_{l=1\ldots n}\left\{D_{u,l}\right\} \tag{11}$$

In this way, we favor the selection of samples placed in areas of the feature space not covered by training samples and avoid choosing samples similar to those already present in the current training set.

Pseudocode 3 synthesizes the proposed strategy based on the distance from the closest training sample.

Pseudocode 3: Active learning based on distance from the closest training sample
Inputs:
L, initial training set.
U, learning set.
n_s, number of samples to add at every iteration of the active learning process.
(1) Compute the Euclidean distances $\mathbf{D}_u \in \Re^n = [D_{u,1}, D_{u,2}, \ldots, D_{u,n}]$ from the n different training samples for each sample $\mathbf{x}_u$ $(u = n+1, n+2, \ldots, n+m)$ of the learning set U.
(2) Identify the training sample $T_{\min,u}$ closest to each sample $\mathbf{x}_u$.
(3) Consider the distance value $D_{\min,u}$ associated with the training sample $T_{\min,u}$.
(4) Set $f_{DAL}(u) = -D_{\min,u}$.
Output:
L'_s, new subset of labeled samples to add in the training set.

3.2. Proposed active learning strategies for SVR

3.2.1. Pool of regressors (SVR-PAL). The first strategy (SVR-PAL) is identical to that presented previously for PLSR. We refer the reader to Section 3.1.1 for more details.

3.2.2. Distance from the support vectors (SVR-DAL). The second method (SVR-DAL) proposed for SVR is very similar to DAL presented previously for PLSR, RR, and KRR. However,

Copyright © 2012 John Wiley & Sons, Ltd.

whereas for DAL we calculate the distances with respect to all training samples, in this case, we consider only the training samples identified as support vectors (SVs) after the regressor training on L. This is motivated by the fact that whereas for DAL all training samples contribute to describing the regression model, for SVR only the SVs are necessary to define the regression function. Moreover, more complex sorting and selection strategies are performed in order to take into account the distribution in the feature space of the samples. First, for each sample $\mathbf{x}_u$ $(u = n+1, n+2, \ldots, n+m)$, the closest support vector $s_{\min,u}$ is identified and the corresponding distance value $D_{\min,u}$ is calculated. Then, we define as $\tilde{\alpha}_u$ the absolute value of the Lagrange multiplier associated with $s_{\min,u}$, that is, $\tilde{\alpha}_u = |\alpha_u - \alpha_u^*|$. We recall that the Lagrange multipliers weight each training sample according to its importance in determining the final solution. The most important training samples are those for which the corresponding Lagrange multipliers are, in absolute terms, equal to the regularization parameter *C*. At this point, the samples of the learning set U are ordered first by function of the value $\tilde{\alpha}_u$ and then by function of the distance value $D_{\min,u}$. Finally, an additional constraint is included in the sample selection step because, at each iteration, it is required that the selected samples cannot share the same closest support vector. In this way, we tend to select most distributed samples possible in the feature space.

Pseudocode 4 synthesizes the proposed method based on the distance from the SVs.

Pseudocode 4: SVR active learning based on distance from the support vectors
Inputs:
L, initial training set.
U, learning set.
n_s, number of samples to add at every iteration of the active learning process.
(1) Identify the *Sn* support vectors of the regressor on the training set L.
(2) Compute the Euclidean distances $\mathbf{D}_u \in \Re^{Sn} = [D_{u,1}, D_{u,2}, \ldots, D_{u,Sn}]$ from the *Sn* different support vectors for each sample $\mathbf{x}_u$ $(u = n+1, n+2, \ldots, n+m)$ of the learning set U.
(3) Identify the support vector $s_{\min,u}$ closest to each sample $\mathbf{x}_u$.
(4) Consider the distance value $D_{\min,u}$ associated with the support vector $s_{\min,u}$.
(5) Consider the absolute value $\tilde{\alpha}_u$ of the Lagrange multiplier associated with the support vector $s_{\min,u}$.
(6) Set $f_{DAL-SVR}(u) = -\tilde{\alpha}_u$, $f_{DSVAL-SVR}(u) = -D_{\min,u}$.
(7) Select first by function of $f_{DAL-SVR}$ and then by function of $f_{DSVAL-SVR}$ if the same value of $f_{DAL-SVR}$ is obtained for successively sorted samples. During the selection, if a sample shares the same closest SV with respect to a previously selected sample, skip it.
Output:
L'_s, new subset of labeled samples to add in the training set.

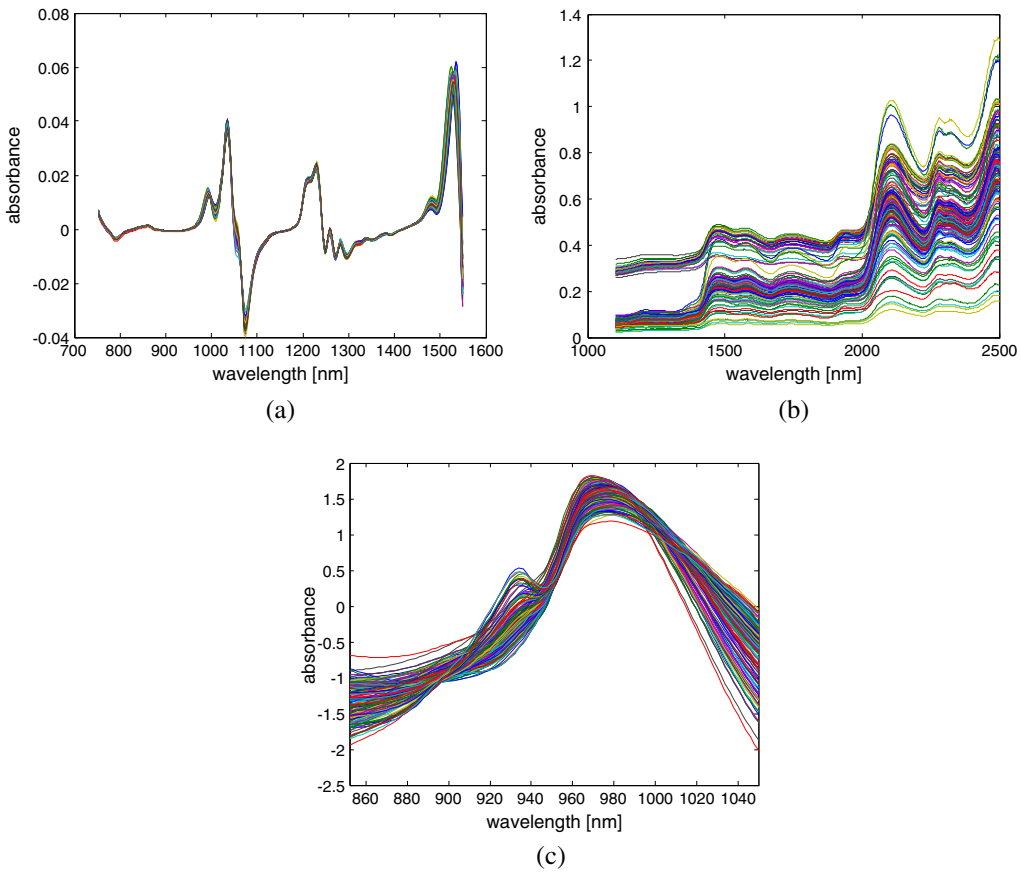


Figure 2. Spectra of the (a) diesel, (b) orange juice, and (c) Tecator data sets.

Copyright © 2012 John Wiley & Sons, Ltd.

4. EXPERIMENTS

4.1. Data set description and experimental setup

In order to validate the proposed active learning methods, we have conducted an experimental study on three real data sets.

The first data set refers to multispectral acquisitions of diesel fuels [28]. It was built by the Southwest Research Institute in order to develop instrumentation to evaluate fuel on battle fields. Along with the spectral acquisitions, different properties are available, such as boiling point at 50% recovery, cetane number, density, freezing temperature, total aromatics, and viscosity. The data set contains only summer fuels, and outliers were removed. In our experiments, we consider one of the most difficult prediction tasks in this data set, that is, the prediction of the cetane number of the fuel. All spectra range from 750 to 1550 nm, discretized into 401 wavelength values. The data set contains 20 high leverage spectra, shown in Figure 2(a), and 225 low leverage spectra, the latter being separated into two subsets labeled a and b. As suggested by the providers of the data, we have built a learning set with the high leverage spectra and a subset of the low leverage spectra (thus yielding 133 spectra). The test set is made up of the low leverage spectra of subset b (gathering 112 spectra).

The second data set deals with the problem of determining sugar (saccharose) concentration in orange juice samples by near-infrared reflectance spectroscopy [29]. The acquisitions consist of 700 spectral variables representing the absorbance (log 1/R) at different wavelengths between 1100 and 2500 nm. The absorbance is defined as log (1/R), where R is the light reflected by the sample surface. In this case, learning and test sets contain 149 and 67 samples, respectively. In Figure 2(b), we show the near-infrared spectra of the orange juice learning set.

The last data set deals with the determination of the fat content in meat samples, analyzed by near-infrared transmittance spectroscopy [30]. Each sample contains finely chopped pure meat with

Table 1. Dataset information and experimental setup for the different data sets

Dataset information				Experimental setup	
Name	# features	# learning samples	# test samples	# initial training samples	# samples added at each iteration
Diesel	401	133	112	33	20
Orange juice	700	149	67	49	20
Tecator	100	172	43	32	20

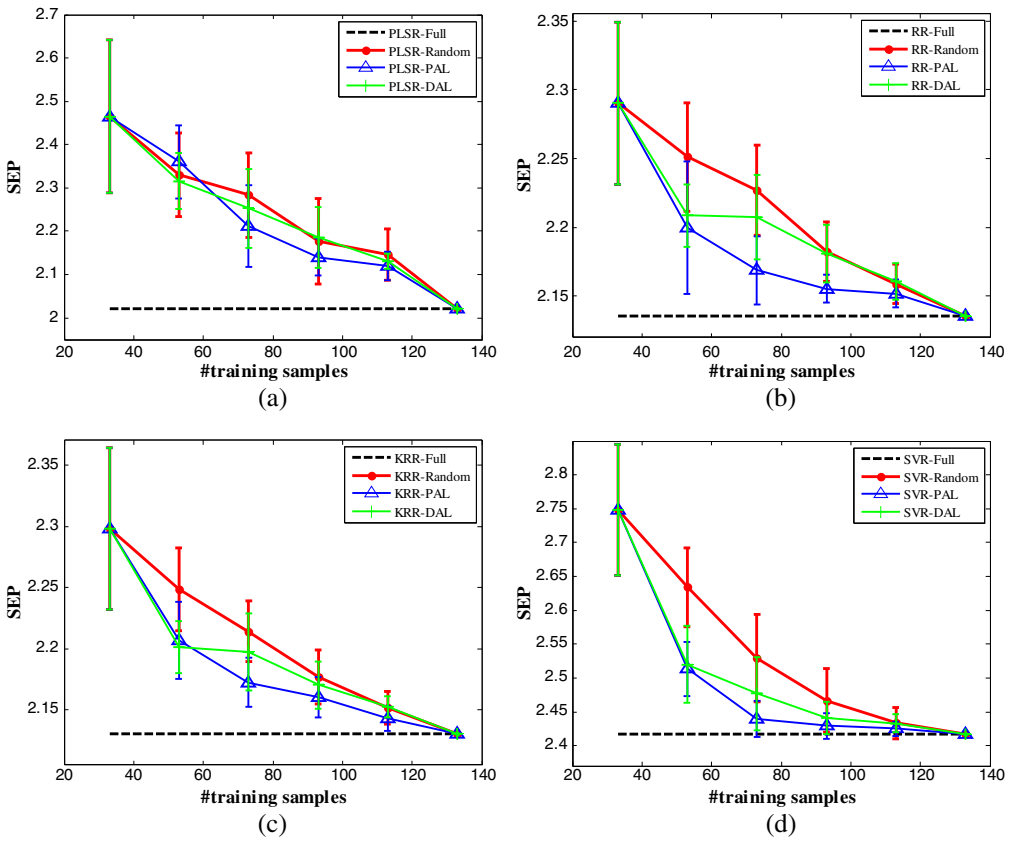


Figure 3. Performances achieved on the diesel data set for (a) PLSR, (b) RR, (c) KRR, and (d) SVR in terms of SEP and standard deviation. Each graph shows the results in function of the number of interactions. All results are averaged over 10 runs of the approaches. (PLSR, RR, KRR, SVR)-Full = full; (PLSR, RR, KRR, SVR)-Random = random; (PLSR, RR, KRR, SVR)-PAL = pool of regressors; (PLSR, RR, KRR)-DAL = distance from the closest training sample in features space; SVR-DAL = distance from the support vectors.

Copyright © 2012 John Wiley & Sons, Ltd.

different moisture, fat and protein contents. Those contents, measured in per cents, are determined by analytic chemistry. The spectra, acquired by the Tecator Infratec Food and Feed Analyzer, records light transmittance through the meat samples at 100 wavelengths in the range between 850 and 1050 nm. The corresponding 100 spectral variables are the absorbance defined by the measured transmittance values. The spectra are normalized according to the standard normal variance method, that is, mean equal to zero and variance equal to one. For this data set, learning and test sets contain 172 and 43 spectra, respectively. The near-infrared spectra of the Tecator learning set is depicted in Figure 2(c).

In all the following experiments, for all data sets, the initial training samples required by the active learning process were selected randomly from the learning set U. For the diesel data set, starting from 33 samples, the active learning algorithms were run until all the learning samples were added to the training set, adding 20 samples at each iteration. Similarly, 20 samples were added at each iteration by starting from 49 and 32 samples for the orange juice and Tecator data sets, respectively. The details of the experimental setup on the different data sets are summarized in Table 1. To yield statistically reliable results, the entire active learning process was run 10 times, each time with a different initial training set. At each run, the initial training samples were chosen in a completely random way.

Linear and nonlinear regressors were also trained on the entire learning set in order to have a reference-training scenario, called "full" training (named PLSR-Full, RR-Full, KRR-Full, and SVR-Full, respectively). On the one hand, the regression results obtained in this way represent a lower boundary for the errors. On the other hand, we expect that the upper error boundary will be given by the completely random selection strategy (named PLSR-Random, RR-Random, KRR-Random, and SVR-Random, respectively). We recall that the purpose of any active learning strategy is to converge to the performance of the "full" training scenario faster than the random selection method.

Regression performances were evaluated on the test sets in terms of the standard error of prediction (SEP):

$$\mathrm{SEP} = \sqrt{\frac{1}{t}\sum_{i=1}^{t}(\hat{y}_i - y_i)^2} \tag{12}$$

where t is the number of test samples.

Concerning the parameter setting, in the case of PLSR, the optimal number of latent variables was estimated in the range [1,20]. For RR, the value of the ridge parameter λ is in the range $[2^{-23}, 2^{-6}]$. For the KRR and SVR, we adopted a Gaussian kernel. This choice is motivated by the generally good prediction accuracy associated with this kernel. For KRR, the value of the kernel width (σ) was chosen in the interval $[2^3, 2^7]$, whereas λ is comprised in $[2^{-23}, 2^{-6}]$. For SVR, the regularization (C) and kernel width (γ) parameters were tuned empirically in the ranges $[2^{-9}, 2^9]$ and $[2^{-11}, 2^3]$, respectively. For all algorithms, parameter values were estimated by cross-validation with five folds. Regarding the number q of regressors for the active learning strategy based on the pool, it was fixed to four in all experiments.

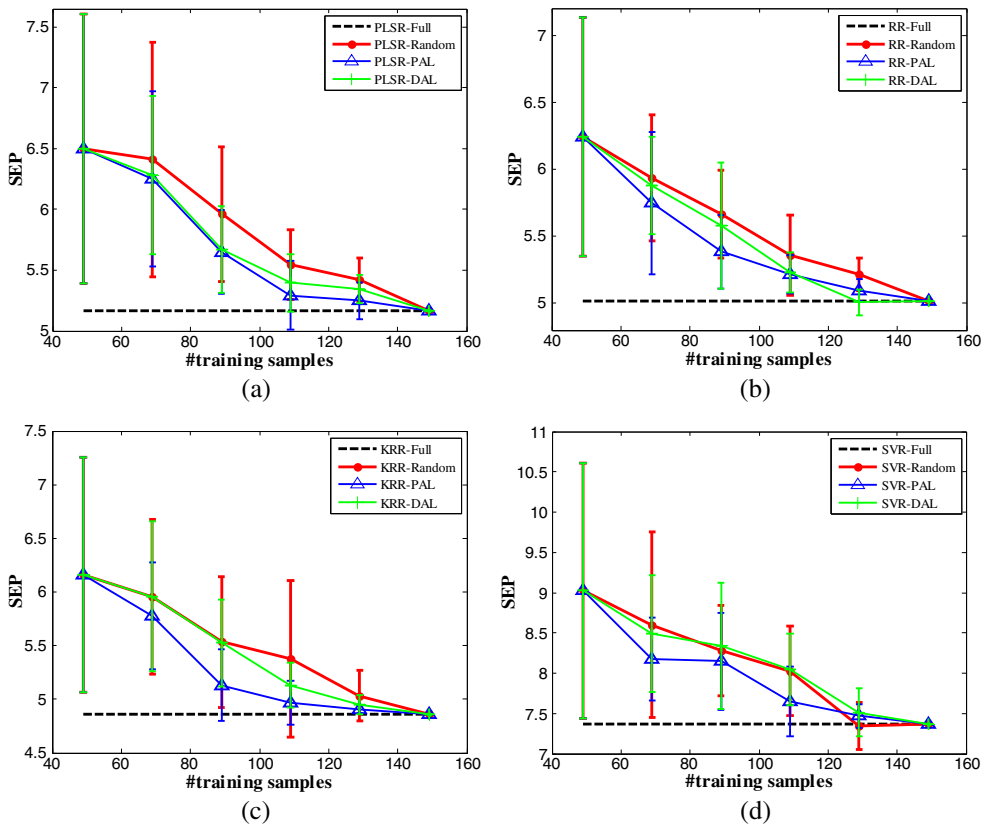


Figure 4. Performances achieved on the orange juice data set for (a) PLSR, (b) RR, (c) KRR, and (d) SVR in terms of SEP and standard deviation. Each graph shows the results in function of the number of interactions. All results are averaged over 10 runs of the approaches. (PLSR, RR, KRR, SVR)-Full = full; (PLSR, RR, KRR, SVR)-Random = random; (PLSR, RR, KRR, SVR)-PAL = pool of regressors; (PLSR, RR, KRR)-DAL = distance from the closest training sample in features space; SVR-DAL = distance from the support vectors.

Copyright © 2012 John Wiley & Sons, Ltd.

4.2. Experimental results

Figures 3–5 report the results obtained for the diesel, orange juice, and Tecator data sets, respectively, by evolving the active learning process. In particular, the graphs refer to (a) PLSR, (b) RR, (c) KRR, and (d) SVR in terms of SEP and standard deviation. First, we note that, before starting the active learning process, poor performances were obtained, both in terms of prediction errors (SEP) and related standard deviations (STD). This result can be expected because of the small number of training samples used to train the regressors, which has also a direct impact on the regression model quality as shown by the strong variability (STD) of the prediction errors. Another expected result is given by the improvement of performances when additional samples are inserted in the training set. This results in graphs with approximately monotonous decreasing behavior of SEP and STD, which tend to converge to the results yielded by the "full" regressors, for which the entire learning set is exploited to train the model. Although such decreasing is verified for both active and random selection, we note that, in general, the active methods allow a faster convergence to the "full" result with respect to the random strategy, both in terms of SEP and STD. In particular, the improvements in terms of STD indicate greater levels of stability in defining the regression model. Whereas for the random selection the entire set of learning samples is necessary to converge for all experiments, in some cases, the active learning process allows complete convergence by using just a subset of the learning set. Moreover, before convergence, the proposed active learning strategies give, in general, an improvement with respect to the random one. This means that similar values of prediction errors can be obtained using a minor quantity of training samples, which implies a reduction in the expert work and a decrease in the computational time necessary to train the regressor. Among the proposed strategies, the PAL method, based on the pool of regressors, yields, in general, better results with respect to those of DAL (based on the distances in the feature space between labeled and unlabeled samples). This is verified for PLSR, RR, KRR, and SVR.

The obtained results are shown in greater detail in Table 2, for the diesel, orange juice, and Tecator data sets, respectively. In particular, we considered the performances obtained when 40 additional samples are inserted in the training set. Therefore, the number of training samples equal to 73, 89, and 72 are considered for the different data sets, respectively. We report the values of SEP and the corresponding STD. The best results are highlighted in bold font. Moreover, for PLSR, we indicate the number of optimal latent variables estimated automatically by cross-validation, whereas for SVR, we show the number of support vectors identified in the training process. As can be seen, the proposed strategies are characterized by better performances with respect to the random method from different points of view. First, better values of accuracies are obtained using the same number of training samples. Then, better values of standard deviations associated with the prediction errors are verified.

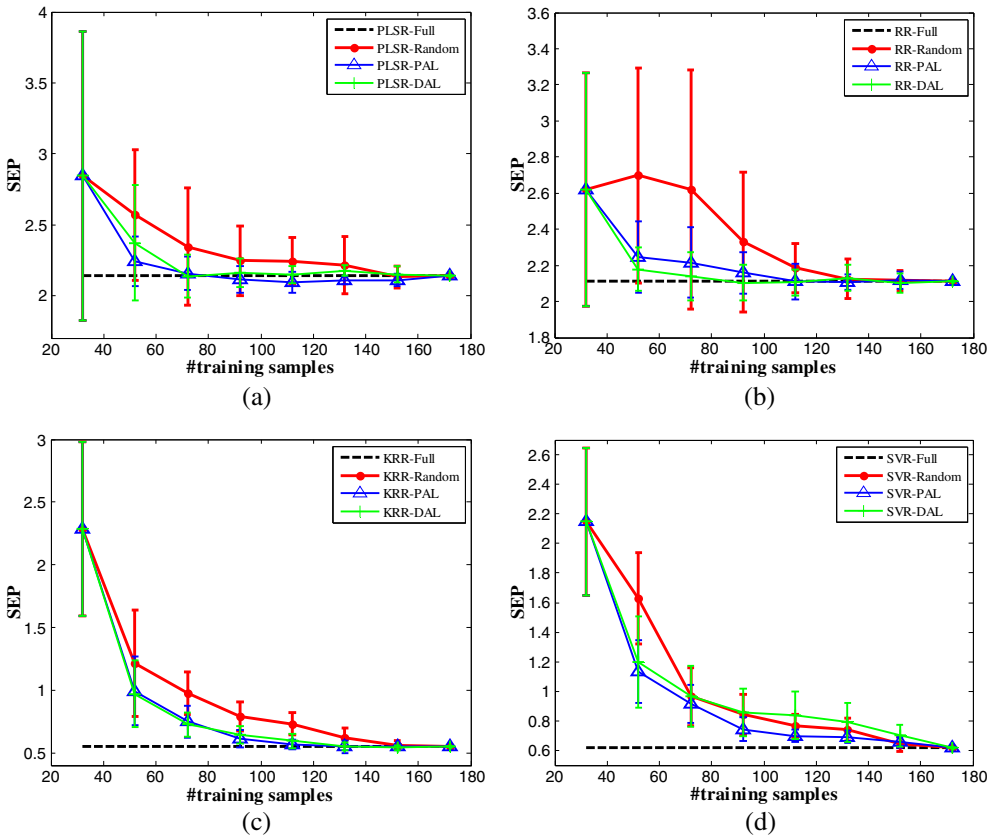


Figure 5. Performances achieved on the Tecator data set for (a) PLSR, (b) RR, (c) KRR, and (d) SVR in terms of SEP and standard deviation. Each graph shows the results in function of the number of interactions. All results are averaged over 10 runs of the approaches. (PLSR, RR, KRR, SVR)-Full = full; (PLSR, RR, KRR, SVR) -Random = random; (PLSR, RR, KRR, SVR)-PAL = pool of regressors; (PLSR, RR, KRR)-DAL = distance from the closest training sample in features space; SVR-DAL = distance from the support vectors.

Copyright © 2012 John Wiley & Sons, Ltd.

Table 2. SEP, Standard Deviation (STD), # latent variables, # support vectors obtained for the PLSR, RR, KRR and the SVR on (a) the diesel, (b) the orange juice, and (c) the Tecator data sets

		linear regression					nonlinear regression				
		PLSR			RR		KRR		SVR		
Method	#training samples	SEP	STD	#latent variables	SEP	STD	SEP	STD	SEP	STD	#support vectors
					(a)						
Full	133	2.0222	-	7	2.1359	-	2.1306	-	2.4171	-	110
Initial	33	2.4651	0.1766	4.4	2.2903	0.0590	2.2978	0.0661	2.7478	0.0968	31.6
Random	73	2.2835	0.09817	4.8	2.2267	0.0328	2.2143	0.0247	2.5296	0.0648	65.5
PAL	73	**2.2118**	0.09371	5.5	**2.1688**	**0.0249**	**2.1725**	**0.0201**	**2.4393**	**0.0262**	68
DAL	73	2.2528	**0.09108**	4.9	2.2077	0.0308	2.1973	0.0313	2.4768	0.0543	66.2
					(b)						
Full	149	5.1688	-	13	5.0137	-	4.8605	-	7.3729	-	142
Initial	49	6.4987	1.1063	9.8	6.2419	0.8930	6.1585	1.0959	9.0254	1.5838	47.7
Random	89	5.9608	0.5542	10.6	5.6648	0.3292	5.5344	0.6124	8.2810	**0.5606**	89
PAL	89	**5.6495**	**0.3431**	13.3	**5.3846**	**0.2762**	**5.1311**	**0.3306**	**8.1491**	0.5995	88.7
DAL	89	5.6693	0.3593	11.6	5.5807	0.4680	5.5251	0.4030	8.3395	0.7867	89
					(c)						
Full	172	2.1377	-	10	2.1137	-	0.5508	-	0.6214	-	151
Initial	32	2.8436	1.0167	4.5	2.6202	0.6451	2.2865	0.6929	2.1485	0.4972	30
Random	72	2.3443	0.4118	9.6	2.6202	0.6601	0.9748	0.1664	0.9646	0.1968	65.6
PAL	72	2.1572	**0.1186**	8.2	2.2151	0.1941	0.7495	0.1264	**0.9133**	**0.1271**	65.8
DAL	72	**2.1351**	0.1519	9.6	**2.1396**	**0.1347**	**0.7252**	**0.0940**	0.9649	0.2059	66.6

5. CONCLUSION

In this work, we have introduced the active learning approach in facing the problem of training sample collection for regression problems related to the estimation of chemical concentrations from spectroscopic data. Starting from a small and suboptimal training set, an iterative process selects, from a set of unlabeled data, the samples considered more significant for the regression process, that is, those able to give smaller prediction errors while minimizing the number of training samples and thus the expert effort and costs for collecting the final training set. In particular, we have proposed two different active learning strategies and tested them for two linear and two nonlinear regression approaches. The first strategy uses a pool of regressors in order to select the samples with the greatest disagreements among the different regressors, while the second relies on the idea of adding samples that are distant from the current training samples. Making an analogy with feature selection methods, we could classify active learning methods into two categories, namely "filter" and "wrapper" methods. The first category of methods select samples independently from the regressor used (it is the case of DAL for PLSR, RR, and KRR), whereas the second category exploits information from the regressor (the case of PAL and SVR-DAL).

The experimental results on three different real data sets show good capabilities of the proposed strategies for selecting significant samples. In general, the proposed methods are characterized by higher performances in terms of both accuracy and stability with respect to a completely random selection strategy. Comparing them, the best active strategy appears to be the one based on the pool of regressors for both linear (PLSR, RR) and nonlinear (KRR, SVR) methods. It is, however, the most computationally demanding because it needs the training of different regressors to build the pool.

Computational cost is the main drawback of the active learning approach. In all cases, it does not go beyond 1 h, including parameter setting. This contained drawback is, however, widely compensated for by the reduction in required training samples, which involves a substantial benefit in terms of time and economic cost incurred by the demanding training sample collection.

Although we focused on PLSR, RR, KRR, and SVR in this paper, the active selection of the training samples could be used in combination with other supervised regression approaches. Moreover, whereas in this work, the initial training set was chosen in a random way, more sophisticated initialization strategies could be envisioned in order to further improve the performances of the active learning process. Finally, other active learning strategies could be considered for SVR in particular, for instance by combining sample distance, pool disagreement and Lagrange multiplier information sources.

Acknowledgements

This research was carried out within the framework of the Averroès Erasmus Mundus program funded by the European Commission. The authors wish to thank Prof. M. Meurens, Université Catholique de Louvain (Belgium), for providing the orange juice data set used in the experiments, and Dr C-C. Chang and Dr C-J. Lin for supplying the software LIBSVM (http://www.csie.ntu.edu.tw/~cjlin/libsvm) used in this research.

REFERENCES

1. Sekulic S, Ward HW, Brannegan D, Stanley E, Evans C, Sciavolino S, Hailey P, Aldridge P. On-line monitoring of powder blend homogeneity by near-infrared spectroscopy. *Anal. Chem.* 1996; **68**: 509–513.
2. Blanco M, Coello J, Eustaquio A, Iturriaga H, Maspoch S. Analytical control of pharmaceutical production steps by near infrared reflectance spectroscopy. *Anal. Chim. Acta* 1999; **392**(2–3): 237–246.

Copyright © 2012 John Wiley & Sons, Ltd.

3. Ozaki Y, Cho R, Ikegaya K, Muraishi S, Kawauchi K. Potential of near-infrared Fourier transform Raman spectroscopy in food analysis. *Appl. Spectrosc.* 1992; **46**: 1503–1507.
4. Blanco M, Coello J, Garcia Fraga JM, Iturriaga H, Maspoch S, Pagès J. Determination of finishing oils in acrylic fibers by near-infrared reflectance spectroscopy. *Analyst* 1999; **122**: 777–781.
5. Geladi P. Some recent trends in the calibration literature. *Chemom. Intell. Lab. Syst.* 2002; **60**: 211–224.
6. Benoudjit N, Melgani F, Bouzgou H. Multiple regression systems for spectrophotometric data analysis. *Chemom. Intell. Lab. Syst.* 2009; **95**: 144–149.
7. Li H, Liang Y, Xu Q. Support vector machines and its applications in chemistry. *Chemom. Intell. Lab. Syst.* 2009; **95**: 188–198.
8. Zhu X. Semi-supervised learning literature survey. Comput. Sci. Tech. Rep. 1530, Univ. Wisconsin-Madison (2005).
9. Gujral P, Amrhein M, Ergon R, Wise BM, Bonvin D. On multivariate calibration with unlabeled data. *J. Chemom* 2011; **25**(8): 456–465.
10. Toher D, Downey G, Murphy TB. Semi-supervised linear discriminant analysis. *J. Chemom* 2011; **25**: 621–630.
11. Settles B. Active learning literature survey. Comput. Sci. Tech. Rep. 1648, Univ. Wisconsin-Madison (2009).
12. Mitra P, Murthy CA, Pal SK. A probabilistic active support vector learning algorithm. *IEEE Trans. Pattern Anal. Mach. Intell.* 2004; **26**(3): 413–418.
13. Pasolli E, Melgani F. Active learning methods for electrocardiographic signal classification. *IEEE Trans. Inform. Technol. Biomed.* 2010; **14**(6): 1405–1416.
14. Pasolli E, Melgani F, Bazi Y. Support vector machine active learning through significance space construction. *IEEE Geosci. Remote Sens. Lett.* 2011; **8**(3): 431–435.
15. Zomer S, Sanchez MDN, Brereton RG, Pavon JLP. Active learning support vector machines for optimal sample selection in classification. *J. Chemom* 2004; **18**(6): 294–305.
16. Cohn D, Ghahramani Z, Jordan M. Active learning with statistical models. *J. Artif. Intell. Res.* 1996; **4**: 129–145.
17. Fukumizu K. Statistical active learning in multilayer perceptrons. *IEEE Trans. Neural Netw.* 2000; **11**(1): 17–26.
18. Castro R, Willett R, Nowak R. Faster rates in regression via active learning. *Adv. Neural Inf. Process. Syst.* 2006; **18**: 179–186.
19. Sugiyama M. Active learning in approximately linear regression based on conditional expectation of generalization error. *The Journal of Machine Learning Research* 2006; **7**: 141–166.
20. Burbidge R, Rowland JJ, King RD. Active learning for regression based on query by committee. *Proceedings of the Intelligent Data Engineering and Automated Learning* 2007; **1**: 209–218.
21. Sugiyama M, RubensN. A batch ensemble approach to active learning with model selection. *Neural Netw.* 2008; **21**(9): 1278–1286.
22. Sugiyama M, Nakajima S. Pool-based active learning in approximate linear regression. *Mach. Learn.* 2009; **75**(3): 249–274.
23. Paisley J, Liao X, Carin L. Active learning and basis selection for kernel-based linear models: a Bayesian perspective. *IEEE Trans. Signal Process.* 2010; **58**(5): 2686–2700.
24. Wold S, Martens H, Wold H. The multivariate calibration problem in chemistry solved by the PLS method. *Matrix Pencils* 1983; **973**: 286–293.
25. Gusnanto A, Pawitan Y, Huang J, Lane B. Variable selection in random calibration of near-infrared instruments: ridge regression and partial least squares regression settings. *J. Chemom.* 2003; **17**(3): 174–185.
26. Cristianini N, Taylor JS. *An introduction to support vector machines and other kernel-based learning methods*. Cambridge University Press: Cambridge, UK, 2000.
27. Smola A, Schölkopf B. A tutorial on support vector regression. *Statistics and computing* 2004; **14**(3): 199–222.
28. Diesel data set available at http://www.eigenvector.com/data/SWRI/.
29. Orange juice data set available at http://www.ucl.ac.be/mlg/.
30. Tecator data set available at http://lib.stat.cmu.edu/datasets/tecator.

 Copyright © 2012 John Wiley & Sons, Ltd.